

Orthogonal complements of adjoint invariant subspaces

Proof of Conjecture 00000006837

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Precise statement and convention

The source pairs invariant subspaces of an operator and its adjoint through orthogonal complements. We prove this standard correspondence for every bounded linear operator A on any real or complex Hilbert space H , without a finite-dimensional assumption. Write A^* for the Hilbert-space adjoint, characterized by $\langle x, A^*y \rangle = \langle Ax, y \rangle$, and define

$$U^\perp = \{y \in H : \langle x, y \rangle = 0 \text{ for every } x \in U\}.$$

An invariant subspace means $A(U) \subseteq U$. In the invariant-subspace lattice of a Hilbert-space operator, subspaces are conventionally closed. We also prove the forward implication for arbitrary linear subspaces, and identify explicitly where closedness is needed for the converse.

Adjoint invariance

Suppose $A(U) \subseteq U$. For $y \in U^\perp$ and $x \in U$,

$$\langle x, A^*y \rangle = \langle Ax, y \rangle = 0,$$

because $Ax \in U$. Thus $A^*(U^\perp) \subseteq U^\perp$. No closedness of U was used in this direction. Conversely, suppose U is closed and $A^*(U^\perp) \subseteq U^\perp$. Apply the result just proved to A^* and $U^\perp$. Using $(A^*)^* = A$ and $(U^\perp)^\perp = U$ gives $A(U) \subseteq U$. Therefore, for every closed U ,

$$A(U) \subseteq U \iff A^*(U^\perp) \subseteq U^\perp.$$

For a nonclosed subspace, the double orthogonal complement is $\overline{U}$, not necessarily U , so this converse is not asserted without closedness.

Full correspondence and order

Every orthogonal complement is closed. Consequently $U \mapsto U^\perp$ maps closed A -invariant subspaces to closed A^* -invariant subspaces. Its inverse is the same orthogonal-complement operation in the other direction, since both double-adjoint and double-complement identities hold. For any closed subspaces U, V ,

$$U \subseteq V \iff V^\perp \subseteq U^\perp.$$

The forward implication follows from the definition of orthogonality; the reverse follows by taking orthogonal complements again. Thus this is an order-reversing bijection between the two closed invariant-subspace lattices. This establishes the orthogonal-complement pairing claimed in both source versions, in the standard bounded Hilbert-space adjoint setting.

Formal verification

Lean is generic over a scalar field with `RCLike` structure and a complete inner-product space, hence covers both $\mathbb{R}$ and $\mathbb{C}$. It uses actual Mathlib continuous linear operators, their adjoints, submodules, orthogonal complements, and topological closedness. The predicate of invariance is the actual condition $\forall x \in U, Ax \in U$. The project proves the arbitrary-subspace implication, the closed subspace equivalence, a bundled equivalence of the two types of closed invariant subspaces, and the inclusion-reversal equivalence. Its axiom audit uses only standard logical axioms; no analytic correspondence or invariance hypothesis is silently assumed as a result.